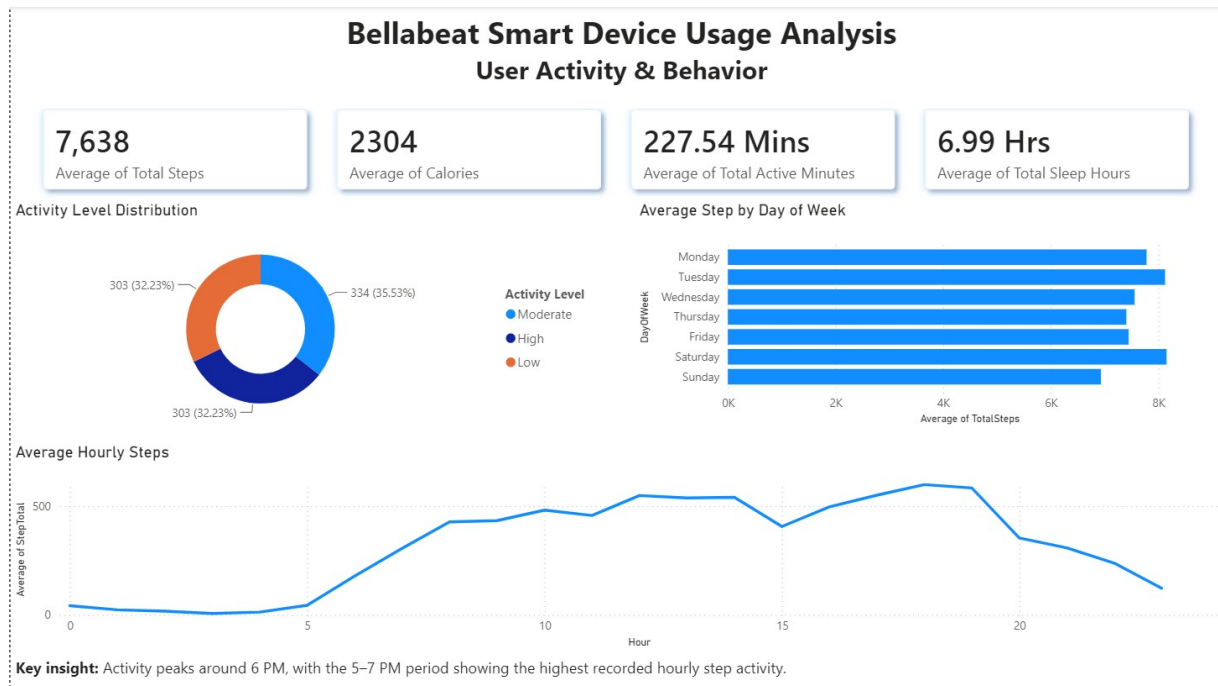


Bellabeat Smart Device Usage Analysis

Marketing Analytics Case Study | Python + Power BI



1. Executive Summary

This case study analyzes Fitbit smart-device usage data to identify patterns in daily activity, sedentary behavior, hourly movement, and sleep, then translates those findings into marketing opportunities for Bellabeat.

Users recorded an average of 7,638 daily steps, 2,303.61 calories, 991.21 sedentary minutes, 227.54 active minutes, and 6.99 hours of sleep. Activity peaked around 6 PM. Low-activity days showed substantially more sedentary time than high-activity days.

2. Business Task

Analyze Fitbit smart-device usage data to identify trends in users' activity, sleep, and daily habits, and use those findings to develop marketing recommendations for Bellabeat.

Key Business Questions

- What are some trends in smart-device usage?
- How could these trends apply to Bellabeat customers?

- How could these trends influence Bellabeat's marketing strategy?

3. Stakeholders

- Bellabeat marketing analytics team
- Bellabeat cofounders and marketing decision-makers
- Product focus: Bellabeat Leaf, with activity, sleep, and stress tracking

4. Data Sources

The Kaggle dataset contained 18 CSV files. Four datasets were selected for the core analysis based on their relevance to the business questions.

- dailyActivity_merged.csv
- sleepDay_merged.csv
- hourlySteps_merged.csv
- hourlyIntensities_merged.csv
- Derived: activity_sleep_analysis.csv

5. Data Audit & Limitations

- Activity and hourly datasets contained 33 unique user IDs; the sleep dataset contained 24.
- The observation period covered approximately April 12 to May 12, 2016.
- The supplied case-study materials describe 30 Fitbit users, while the supplied activity/hourly files contain 33 unique IDs; this discrepancy was retained as a data-quality/documentation issue.
- Not every user has observations for every day, and sleep coverage is more limited than activity coverage.
- The Fitbit sample is a proxy and should not be generalized to all Bellabeat customers.
- Correlation indicates association, not causation.

6. Data Cleaning & Processing

- Loaded and inspected the selected CSV files using pandas.
- Checked dimensions, columns, data types, unique users, missing values, and duplicates.
- Converted date fields to datetime format and created DayOfWeek fields.

- Created TotalSleepHours and TimeInBedHours.
- Created TotalActiveMinutes from very active, fairly active, and lightly active minutes.
- Created Low, Moderate, and High activity categories from daily steps.
- Removed three exact duplicate sleep records.
- Merged activity and sleep observations by user ID and date for relationship analysis.
- Exported cleaned datasets for Power BI.

7. Exploratory Data Analysis

7.1 Overall KPIs

Metric	Result
Average Daily Steps	7,637.91
Average Daily Calories	2,303.61
Average Sedentary Minutes	991.21
Average Active Minutes	227.54
Average Sleep Duration	6.99 hours
Average Time in Bed	7.64 hours
Peak Activity Hour	18:00 / 6 PM

7.2 Activity-Level Distribution

- Moderate: 35.53%
- High: 32.23%
- Low: 32.23%

7.3 Activity Level vs. Behavior

Level	Avg Steps	Sedentary Min	Active Min	Calories
High	13,337.40	888.66	313.76	2,743.58
Moderate	7,465.78	956.09	271.81	2,355.16
Low	2,128.17	1,132.47	92.52	1,806.81

Low-activity days recorded about 244 more sedentary minutes than high-activity days.

7.4 Weekday Activity

- Monday: 7,780.87 steps
- Tuesday: 8,125.01
- Wednesday: 7,559.37
- Thursday: 7,405.84
- Friday: 7,448.23
- Saturday: 8,152.98
- Sunday: 6,933.23

Saturday was the most active day and Sunday the least active; the difference was approximately 1,220 steps.

7.5 Hourly Activity

Average hourly steps peaked at 18:00 (approximately 599), followed by 19:00 (approximately 583) and 17:00 (approximately 550). The 5–7 PM period was the strongest observed activity window.

7.6 Sleep Patterns

- Highest average sleep: Sunday, 7.55 hours.
- Lowest average sleep: Thursday, 6.69 hours.
- Difference between highest and lowest day: approximately 52 minutes.

7.7 Steps vs. Sleep

The Pearson correlation between daily steps and sleep duration was -0.19, indicating a weak negative linear association and little meaningful linear relationship in this sample.

8. Key Insights

- Users recorded substantial sedentary time, averaging approximately 991 minutes per day.
- Low-activity days had far fewer steps and considerably more sedentary time than high-activity days.
- Activity peaked around 6 PM, with the 5–7 PM period showing the strongest hourly activity.
- Saturday was the most active day and Sunday the least active by average steps.
- Sleep varied across the week, with Sunday highest and Thursday lowest.
- Steps and sleep showed only a weak negative association ($r = -0.19$).

9. Marketing Recommendations

Recommendation 1 — Time-Based Evening Activity Engagement

Use the observed 5–7 PM activity window to test personalized evening engagement in the Bellabeat app, such as progress summaries, activity prompts, or wellness content. Treat 6 PM as a testable opportunity rather than a universal optimal time.

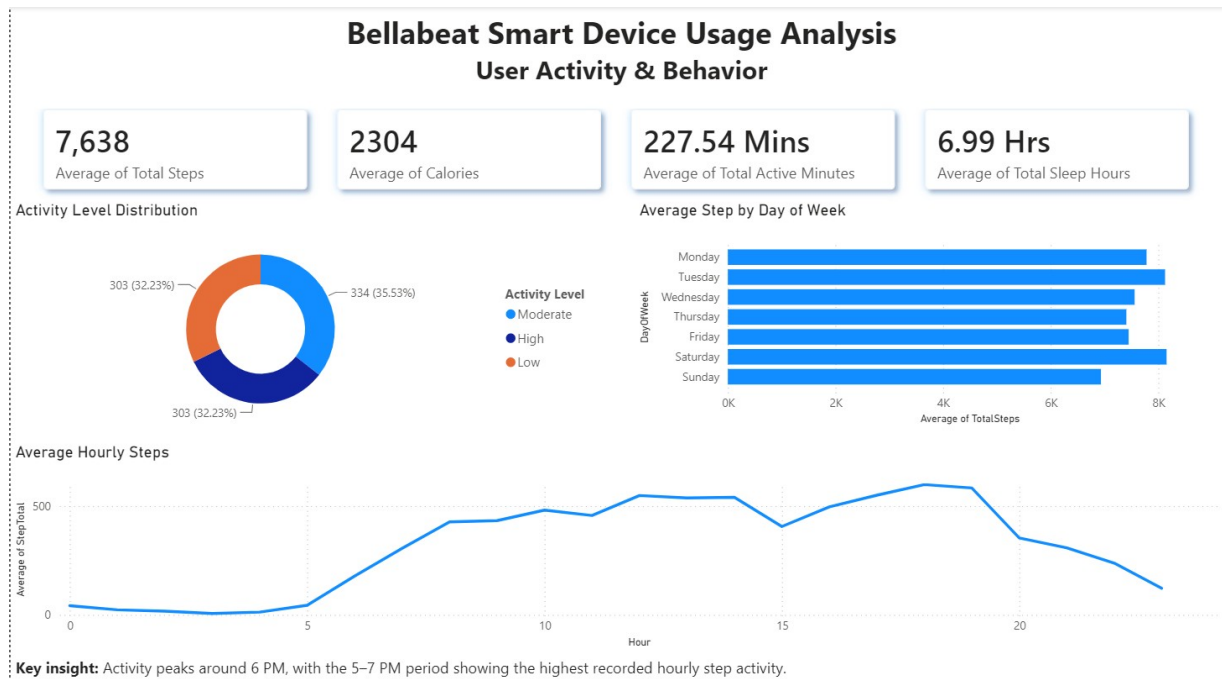
Recommendation 2 — Target Low-Activity / High-Sedentary Behavior

Use activity patterns to segment users and provide more relevant movement-oriented content to users who repeatedly record low activity and high sedentary time.

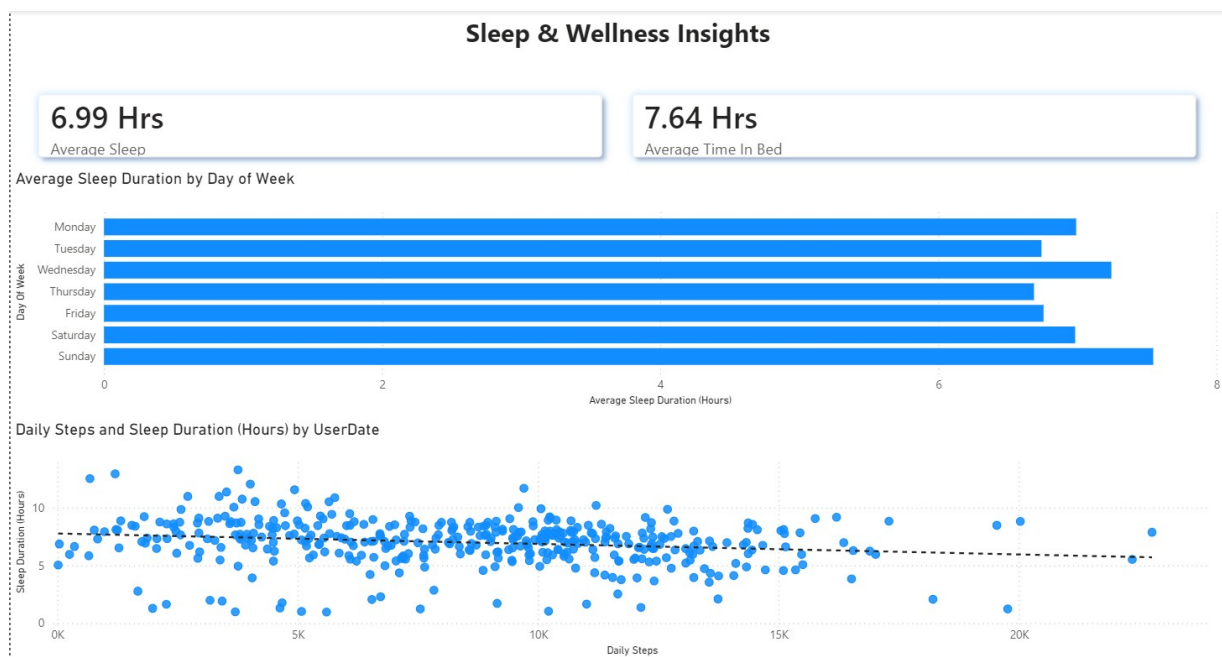
Recommendation 3 — Weekday Sleep & Wellness Engagement

Use observed weekday variation in sleep as a basis for sleep-awareness content and personalized wellness engagement, particularly around lower-sleep parts of the week.

10. Power BI Dashboard



Page 1 — User Activity & Behavior: KPI cards, activity-level distribution, daily steps by weekday, and hourly steps.



Page 2 — Sleep & Wellness Insights: sleep KPIs, sleep by weekday, and daily steps versus sleep duration.

11. Tools Used

- Python
- pandas
- NumPy
- Matplotlib
- Power BI
- DAX

12. Conclusion

The analysis identifies clear differences in activity behavior, a strong evening activity window, and meaningful variation in sleep across the week. The strongest marketing opportunities are personalized evening engagement, segmentation of low-activity/high-sedentary users, and broader sleep and wellness engagement. These recommendations should be validated with Bellabeat-specific customer data before broad deployment.

13. Project Deliverables

- Raw stakeholder CSV files
- Cleaned CSV datasets
- Python/Jupyter analysis
- Power BI dashboard (.pbix)
- Final case-study report
- GitHub README